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PSSF: A Phase-Space and State-Space Fusion Method for EEG Source Localization and Feature Decoding

Zheng Hu[1], Xiaoqi Luo[1], Xu Shao[2], Yunyuan Gao[1,*], and Yanhua Qin[2,*]

[1]School of Automation, Hangzhou Dianzi University, Hangzhou, China [2]Department of Psychiatry, Sir Run Run Shaw Hospital, Zhejiang University School of Medicine, China [*]Authors to whom any correspondence should be addressed.

E-mail: gyy@hdu.edu.cn;21118323@zju.edu.cn

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Abstract

Objective. Scalp electroencephalographic (EEG) signals are affected by volume-conduction-induced mixing, whereas source-localized time series contain nonlinear dynamics together with redundant activity from task-irrelevant regions of interest (ROIs). This study aims to develop an EEG source-localization and feature-decoding method that jointly exploits source-space information and nonlinear dynamics. **Approach.** We propose PSSF, a phase-space and state-space fusion method for EEG source localization and feature decoding. Scalp EEG signals are projected into cortical source space using standardized low-resolution electromagnetic tomography to obtain region-of-interest (ROI)-level source time series. Phase-space reconstruction is then performed through time-delay embedding, and approximate entropy, fuzzy entropy, and permutation entropy are combined to characterize nonlinear source dynamics. In parallel, a state-space model performs sparse source estimation and Bayesian information criterion-based model selection to identify active ROIs. Features retained according to the active-ROI constraints are further ranked using ReliefF, and the selected feature subset is evaluated using a linear support vector machine. **Main results.** PSSF achieved mean classification accuracies of 77.81% on the binary motor-imagery task from BCI Competition IV 2a and 98.35% on the self-collected adult attention-deficit/hyperactivity disorder dataset. In the ablation analysis, the mean classification accuracies obtained using fuzzy entropy and hybrid entropy increased by 6.98 and 5.72 percentage points, respectively, after phase-space reconstruction and state-space-based ROI screening were jointly introduced. Phase-space trajectories, t-SNE distributions, and inter-ROI correlation patterns further revealed between-class dynamical differences and a more concentrated organization in task-related sensorimotor regions. **Significance.** PSSF establishes a connection between source-space activity screening and nonlinear dynamical feature extraction. The results indicate that combining active-ROI constraints with phase-space complexity representations can improve EEG feature discrimination while providing complementary spatial and dynamical information for physiological interpretation.

1 Introduction

Brain-computer interfaces (BCIs) establish direct communication pathways between the brain and external devices, with the key objective of decoding neural intentions from high-dimensional and nonstationary electroencephalographic (EEG) signals[1]. Scalp EEG is a surface projection of intracranial neuronal population activity after attenuation and mixing through the volume conductor. Conventional analysis therefore faces limited spatial resolution and insufficient characterization of temporal dynamics. EEG source localization maps scalp recordings into cortical source space by solving the electromagnetic inverse problem, providing anatomically meaningful spatial information for subsequent neural decoding[2]. However, source-localization outputs are typically high-dimensional source time series. Extracting their latent nonlinear dynamics while reducing redundant information from task-irrelevant ROIs remains a key challenge for BCI recognition performance and interpretability[3].

Existing studies have improved EEG source estimation through inverse-problem formulation and spatiotemporal modeling. Jiao et al.[4] developed an attention-based deep-learning framework that enhanced inverse-problem solving through spatial constraints. Hamid et al.[5] proposed a regional spatiotemporal Kalman filter (RSTKF) based on a state-space model to characterize region-specific dynamics and obtain focused estimates of deep-brain activity. These studies established a basis for source-space analysis, but the nonlinear dynamics contained in source time series after localization remain underused.

Phase-space reconstruction (PSR) provides an effective approach for analyzing the nonlinear dynamics of source time series. Based on nonlinear dynamics and chaos theory, PSR maps a one-dimensional sequence into a high-dimensional space through time-delay embedding and can recover latent attractor structures and dynamical trajectories under appropriate conditions[6]. Chen et al.[7] extracted nonlinear features through amplitude-frequency analysis in reconstructed phase space. Dovedi et al.[8] combined multivariate variational mode decomposition with PSR to characterize phase-space dynamics under different oscillatory modes. Kumar and Kasthuri[9] and Skaria et al.[10] applied PSR to epileptic EEG classification and chaotic-pattern analysis, respectively. Lu et al.[11] further used phase-amplitude coupling to construct functional brain networks and characterize changes in neural interactions across seizure stages. Collectively, these studies show that nonlinear dynamical representations provide information complementary to conventional time- and frequency-domain EEG features.

In addition to temporal dynamics, the spatial distribution of task-related source activity is an important basis for EEG feature analysis. A state-space model (SSM) describes the evolution of latent neural states through state and observation equations and can support dynamic modeling and spatial activity estimation. Wang et al.[12] and Shen et al.[13] applied SSMs to seizure recognition and sleep-stage classification, respectively. Cheung et al.[14] combined an SSM with expectation-maximization to estimate multivariate autoregressive parameters, spatial activity components, and EEG noise covariance. Overall, source-space activity estimation and nonlinear dynamical characterization have commonly been treated as separate analytical processes, and active-ROI information has not been fully incorporated into the selection of dynamical features. To address this gap, PSSF first maps preprocessed EEG into source space and extracts nonlinear dynamical features using PSR and hybrid entropy. In parallel, an SSM identifies active ROIs, which are used as spatial constraints to retain the corresponding features. This design links active-ROI identification with dynamical feature selection and enables coordinated analysis of source-space and nonlinear dynamical information.

The main contributions of this study are as follows:

(1) PSSF is proposed as a method for EEG feature decoding after source localization by integrating a state-space model (SSM) with phase-space reconstruction (PSR). It combines the source-activity modeling capability of the SSM with the nonlinear dynamical representation capability of PSR.

(2) A state-space-constrained phase-space feature-selection mechanism is developed. Active ROIs identified by the SSM are used to screen PSR-derived hybrid entropy features at the ROI level, and ReliefF is subsequently applied to reduce feature redundancy.

(3) PSSF is evaluated on a public motor imagery dataset and a self-collected adult ADHD dataset. Classification, ablation, and visualization results demonstrate favorable feature-decoding performance and reveal between-class dynamical differences and task-related spatial patterns.

2 Related Work

2.1 Phase-Space Reconstruction of EEG Signals

Phase-space reconstruction (PSR) is an analytical method grounded in nonlinear dynamics and chaos theory. Its central idea is to reconstruct the evolution of an underlying dynamical system in a high-dimensional space through time-delay embedding of a scalar time series, thereby revealing latent dynamical structures[15]. EEG activity arises from complex nonlinear interactions among large neuronal populations and exhibits marked nonlinearity and nonstationarity. PSR can therefore extract dynamical information that is difficult to observe directly in the time or frequency domain and provide a basis for subsequent complexity-feature construction and classification[16].

PSR is primarily based on Takens' embedding theorem[6]. The theorem states that, for an unknown deterministic dynamical system, a scalar observation time series can be used to reconstruct a phase-space representation that is topologically equivalent to the original system, provided that an appropriate embedding dimension is used. Analysis of the geometric structure and dynamic properties of the reconstructed trajectories can reveal hidden attractor structures, trajectory complexity, and system stability.

In EEG analysis, PSR can characterize the temporal dynamics of brain activity. Because EEG reflects the continuous temporal evolution of neuronal population activity, its dynamics can be regarded as a projection of a complex high-dimensional system onto the observation space. PSR expands a one-dimensional time series into multidimensional state vectors, allowing a more comprehensive description of dynamic changes across temporal scales. The phase-space representation also enhances the characterization of nonlinear structures, making dynamical patterns that are difficult to observe directly in the time or frequency domain more explicit in the reconstructed space.

2.2 sLORETA-Based EEG Source Localization

EEG source localization maps scalp signals into cortical source space by solving the electromagnetic inverse problem. In general, the EEG generation process can be described using the following linear model[17]:

$$Y = LS + \varepsilon. \tag{1}$$

where $Y \in \mathbb{R}^{N\times T}$ denotes the EEG signals recorded at the scalp electrodes, N is the number of electrodes, and T is the number of time samples; $S \in \mathbb{R}^{M\times T}$ denotes the cortical source signals, with M being the number of candidate neural sources; $L \in \mathbb{R}^{N\times M}$ is the lead-field matrix, which describes the potential response generated at each electrode by a unit current source; and ε denotes measurement noise, which is generally assumed to be zero-mean Gaussian white noise.

Because the number of scalp electrodes is typically much smaller than the number of candidate sources, the EEG inverse problem is underdetermined and ill-posed, requiring appropriate constraints to obtain stable source estimates. Low-resolution electromagnetic tomography (LORETA) estimates distributed source activity using a Laplacian smoothness constraint based on the assumption of spatial continuity, but its results may still be affected by depth bias[18]. To mitigate this issue, standardized low-resolution brain electromagnetic tomography (sLORETA) introduces statistical standardization into LORETA[19]. Specifically, the LORETA estimates are normalized using the diagonal elements of the source current density covariance matrix, reducing scale differences across source locations.

$$\mathbf{J}_{\text{sLORETA}} = \frac{\mathbf{J}_{\text{LORETA}}}{\sqrt{\text{diag}(\mathbf{C}_J)}}. \tag{2}$$

Here, $\text{diag}(C_J)$ denotes the diagonal elements of the covariance matrix C_J and represents the variance of the current-density estimate at each source point. Normalizing different source points by the standard deviation of their estimates reduces inter-location scale differences and depth bias, thereby improving the comparability of source activity across ROIs. Under ideal single-source conditions, sLORETA has the theoretical property of zero localization error and has therefore been widely used for distributed EEG source localization.

2.3 Complexity-Entropy Feature Extraction

EEG signals exhibit marked nonlinear, nonstationary, and complex dynamics. Complexity-entropy methods quantify the intrinsic complexity of neural activity from perspectives such as sequence irregularity, pattern variation, and dynamical stability, and are therefore commonly used to capture EEG dynamics that cannot be adequately represented by conventional time- or frequency-domain features. Common measures include approximate entropy, permutation entropy, and fuzzy entropy. Approximate entropy quantifies the persistence of similar patterns in a time series and reflects signal complexity and unpredictability[20]. Permutation entropy evaluates complexity according to the ordinal patterns of local sequences and their probability distribution, offering high computational efficiency and low sensitivity to amplitude variations[21]. Fuzzy entropy measures pattern similarity using a continuous fuzzy membership function and reduces the sensitivity of the entropy estimate to threshold selection[22]. These three measures characterize the dynamic complexity of source time series from complementary perspectives and provide complementary information for subsequent fused-feature construction.

3 Methods

This section describes the overall workflow of PSSF. Preprocessed EEG is first subjected to sub-band source localization to obtain ROI-level cortical source time series. Time-delay embedding is then used to reconstruct phase-space trajectories, from which a hybrid descriptor comprising approximate, fuzzy, and permutation entropy is extracted. In parallel, a state-space branch identifies active ROIs through sparse source estimation and BIC-based model selection. The dynamical features corresponding to the selected ROIs are retained and further ranked using ReliefF before being evaluated with a common classifier. The overall workflow is shown in figure 1.

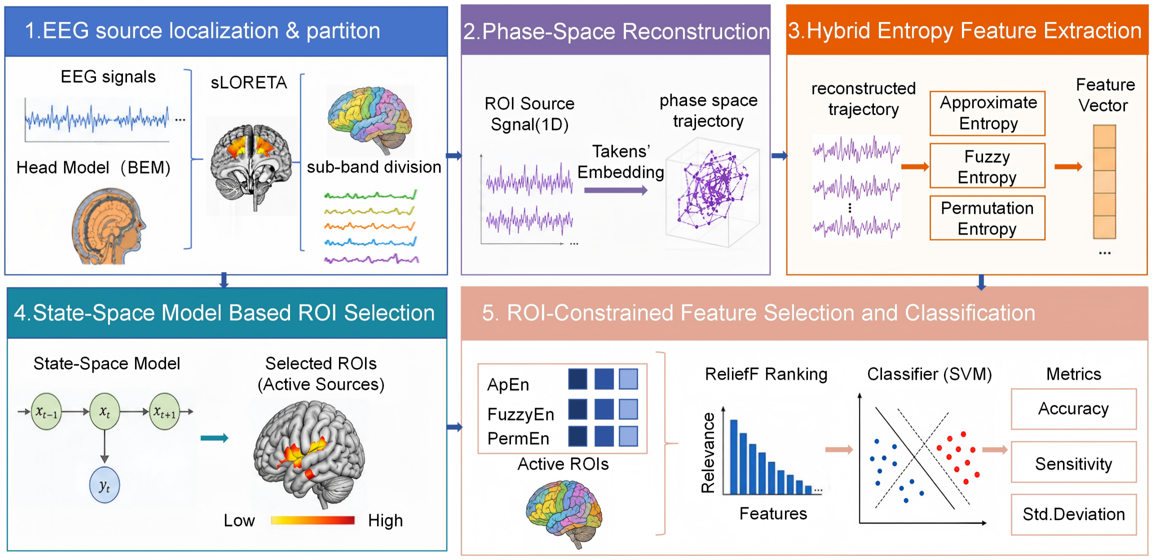


Figure 1. Overall workflow of the PSSF method. (1) EEG source localization and sub-band partitioning; (2) phase-space reconstruction of ROI-level source signals; (3) hybrid entropy feature extraction; (4) state-space-model-based active-ROI selection; and (5) ROI-constrained ReliefF feature selection and SVM classification.

3.1 *Datasets and Preprocessing*

3.1.1 BCI Motor Imagery Dataset The motor imagery dataset used in this study was obtained from BCI Competition IV 2a of the BCI Competition 2008[23]. The dataset was jointly collected by the Institute for Knowledge Discovery and the Institute for Human-Computer Interfaces at Graz University of Technology, Austria, to evaluate multiclass motor imagery EEG classification algorithms. It contains EEG recordings from nine healthy participants, labeled A01–A09.

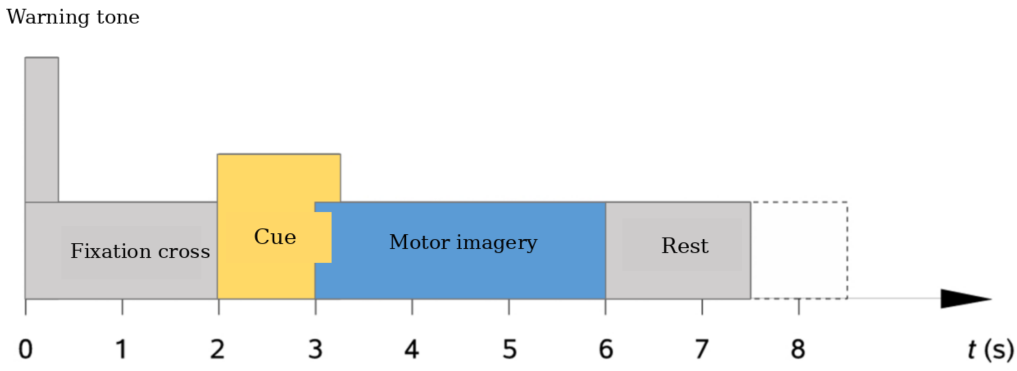


Figure 2. Experimental timeline of a single trial in the BCI Competition IV 2a dataset. A fixation cross and a brief warning tone were presented at 0 s, the task cue appeared at 2 s, and the participant performed the indicated motor imagery until 6 s, followed by a rest period.

The experimental paradigm comprised four cue-based motor imagery tasks: left hand (class 1), right hand (class 2), both feet (class 3), and tongue (class 4). Each participant completed 288 trials, and the timeline of a single trial is shown in figure 2. At t = 0 s, a fixation cross appeared together with a brief acoustic warning tone. At t = 2 s, an arrow pointing in one of four directions was presented for 1.25 s as the task cue. Participants then performed the corresponding motor imagery until the fixation cross disappeared at t = 6 s. No feedback was provided. Artifact-contaminated trials were identified by visual inspection, and approximately 5 min of electrooculography (EOG) data under eyes-open, eyes-closed, and eye-movement conditions were recorded at the beginning of each session to assist in estimating EOG effects.

Only the left- and right-hand motor imagery tasks were analyzed in this study. A 3-s EEG segment was extracted from each trial for subsequent processing and analysis. The raw EEG signals were first band-pass filtered between 0.5 and 40 Hz using a Butterworth filter to preserve motor-imagery-related alpha- and beta-band activity while suppressing low-frequency drift and high-frequency noise. Independent component analysis (ICA) was then performed in EEGLAB. Ocular, jaw-muscle, and other artifact components identified by the ICLabel automated classifier were removed, and segments with excessively large standard deviations were excluded to improve signal quality.

3.1.2 Adult ADHD Dataset The adult attention-deficit/hyperactivity disorder (ADHD) dataset was jointly collected by our research team and Sir Run Run Shaw Hospital, Zhejiang University School of Medicine, to investigate EEG responses of adults with ADHD during attention tasks. The experiment included visual and auditory oddball paradigms designed to examine attention allocation and cognitive processing across sensory modalities. The paradigm is illustrated in figure 3.

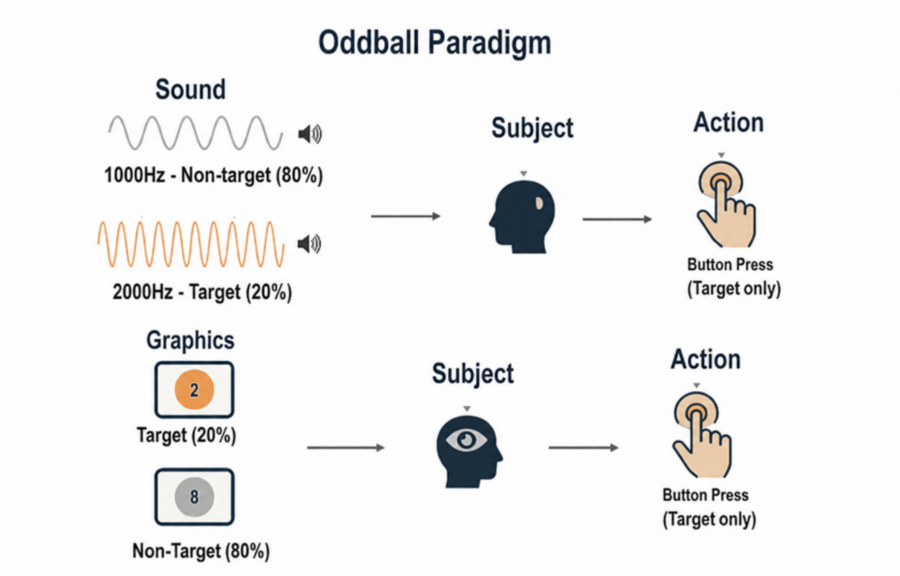


Figure 3. Schematic illustration of the auditory and visual oddball paradigms used in the adult ADHD dataset. In both paradigms, target and non-target stimuli were presented with probabilities of 20% and 80%, respectively, and participants were instructed to respond only to target stimuli.

The paradigm included visual and auditory oddball tasks. In the visual task, an image of the digit ”8” served as the non-target stimulus, whereas an image of the digit ”2” served as the target stimulus. Target and non-target probabilities were 20% and 80%, respectively. Each stimulus lasted 100 ms, and the interstimulus interval was randomly varied from 800 to 1200 ms. A total of 100 target trials were included. In the auditory task, 1000 Hz and 2000 Hz pure tones served as the non-target and target stimuli, respectively. Both tones lasted 200 ms and were delivered binaurally at 75 dB sound pressure level. The target probability was also 20%, and the task included 40 target trials. In both paradigms, participants were instructed to press a response button with the index finger of their dominant hand when a target appeared and to withhold responses to non-target stimuli.

EEG data were recorded using a Neuroscan 64-channel system, with electrodes positioned according to the international 10-10 system. The recording configuration included 64 EEG channels, vertical electrooculography (VEO), horizontal electrooculography (HEO), and a trigger channel, yielding 67 channels in total. The ground electrode was placed at Fz, and signals were referenced to the bilateral mastoids. All electrode impedances were maintained below 10 kOhm, and the sampling rate was 1000 Hz.

The data used in this study underwent standardized preprocessing, including 1–30 Hz band-pass filtering, re-referencing to the bilateral mastoids, bad-channel handling, and independent-component-analysis-based artifact removal. The final dataset comprised 22 auditory recordings and 20 visual recordings, each with 64 EEG channels sampled at 1000 Hz. Visual oddball recordings from four ADHD–HC participant pairs (eight participants in total) were selected for the subsequent PSSF feature-extraction and classification analyses. The four paired datasets are denoted A01–A04.

The study protocol for the self-collected adult ADHD dataset was approved by the Ethics Committee of Sir Run Run Shaw Hospital, Zhejiang University School of Medicine on 20 September 2024 (protocol ID: 2024-2513-01). The study was conducted in accordance with the Declaration of Helsinki and local statutory requirements. Written informed consent was obtained from all participants before data collection.

3.2 Extraction of Band-Specific and ROI-Level EEG Source Time Series

After standardized preprocessing, the ICBM 152 template was adopted as the common head model for all participants and matched to the corresponding EEG channels[24]. The forward model was computed using OpenMEEG, and a realistic three-layer boundary element model (BEM) comprising the cortex, skull, and scalp was constructed. The inverse problem was solved using sLORETA to obtain cortical source estimates. To reduce computational complexity and improve interpretability, high-resolution cortical source points were grouped into 28 regions of interest (ROIs) according to the Brodmann atlas. Figure 4 shows the anatomical distribution of these ROIs. After source localization, source points were assigned to their corresponding ROIs according to anatomical location, and ROI-level source time series were generated as inputs to subsequent phase-space reconstruction, complexity-feature extraction, and active-ROI selection.

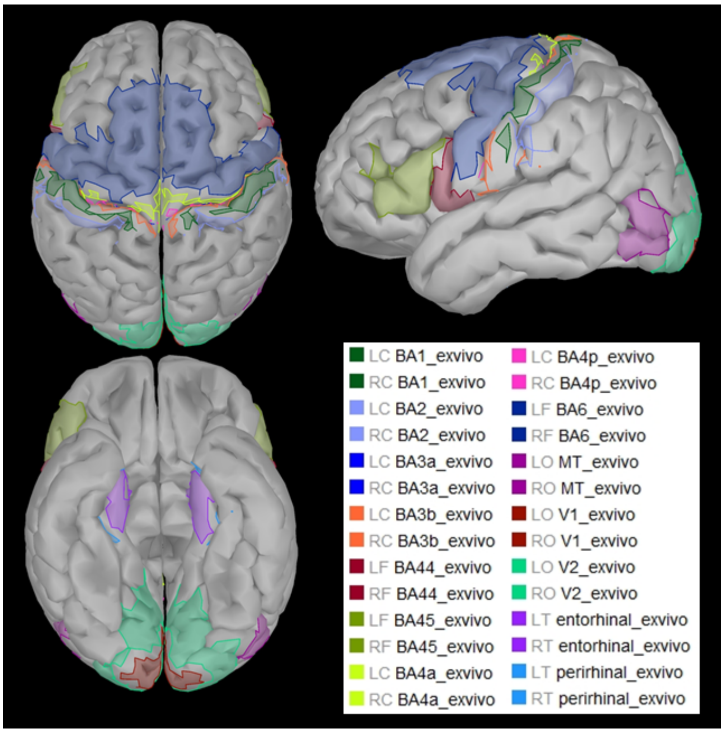


Figure 4. Spatial distribution of the 28 Brodmann-based ROIs used to aggregate cortical source estimates into ROI-level source time series.

Because EEG contains frequency-specific neural oscillations, the signals were further divided into sub-bands[25]. The 8–40 Hz range was partitioned into eight 4-Hz-wide bands. This design retains sufficient oscillatory cycles for stable feature computation while improving the distinguishability of frequency-specific contributions to task classification. The signal partitioning before source localization, the extraction of source-space signals, and the corresponding dimensional transformations are described below.

Let the preprocessed EEG signal in the scalp sensor space be represented as

$$X = [x(1), x(2), \ldots, x(T)] \in \mathbb{R}^{N_c \times T}. \tag{3}$$

where N_c denotes the number of EEG channels (i.e., scalp electrodes), and T is the number of time samples per channel. $X_i(t)$ denotes the voltage recorded at the i-th channel at time t (i = 1, ..., N_c; t = 1, ..., T).

To extract neural oscillations in different frequency ranges, X is band-pass filtered. The frequency range of the f-th sub-band is defined as $\mathcal{B}_f = [8 + 4(f-1),\, 12 + 4(f-1)]$ Hz, $f = 1, 2, \ldots, 8$. Thus, the input before source localization is partitioned into eight independent data blocks, each retaining the original spatial and temporal dimensions of the sensor-space signals.

To improve physiological interpretability, the cortex is divided into 28 ROIs according to the Brodmann atlas. After source localization, source points are grouped into their corresponding ROIs according to anatomical location, and the ROI source time series are organized using ROIs as channels. For the f-th sub-band, the resulting ROI source signal is represented as $Y_f \in \mathbb{R}^{N_r \times T \times N_{\text{trial}}}$, where N_r denotes the number of ROIs, T is the number of time samples per trial, and N_{trial} is the number of trials. The resulting EEG source time series retain frequency-band, ROI, and temporal information and are used as inputs for subsequent PSR, feature extraction, and classification.

3.3 Phase-Space Reconstruction of EEG Source Time Series

According to Takens' embedding theorem, a one-dimensional sequence can be mapped into a high-dimensional phase space by constructing time-delayed embedding vectors. For each ROI and trial, the maximum embedding dimension was set to 8, the maximum time delay to 15, and the false-nearest-neighbor threshold to 10%. The algorithm searched for a time delay and minimum embedding dimension at which the false-nearest-neighbor ratio fell below the threshold and stabilized[26, 27]. The reconstructed trajectory is given by

$$Y_{c,r} \in \mathbb{R}^{L^*_{c,r} \times d^*_{c,r}}, \qquad L^*_{c,r} = T - (d^*_{c,r} - 1)\tau^*_{c,r}. \tag{4}$$

Because the embedding parameters vary across ROIs and trials, the reconstructed trajectories have different lengths. To obtain a uniform representation, the minimum valid trajectory length was identified and all trajectories were truncated accordingly. The Euclidean norm of each state vector was then calculated to convert the multidimensional trajectory into a one-dimensional sequence while retaining changes in trajectory magnitude. The transformed sequences from all ROIs and trials were organized into a three-dimensional array $\mathbf{Y}_{\text{PSR}} \in \mathbb{R}^{L_{\min} \times N_r \times N_{\text{trial}}}$, where $L_{\min}$ denotes the unified sequence length, N_r is the number of ROIs, and N_{trial} is the number of trials. This array represents the EEG source signals after phase-space reconstruction and norm transformation and is used for subsequent feature extraction and classification.

3.4 Hybrid Entropy Feature Extraction from EEG Source Signals

Feature extraction was performed to convert the high-dimensional evolution of the reconstructed trajectories into representations suitable for machine-learning classifiers. EEG exhibits pronounced nonlinearity, nonstationarity, and complex dynamics, which may not be fully described by conventional time- or frequency-domain features alone[28]. Nonlinear complexity measures were therefore extracted from the reconstructed phase-space trajectories to characterize the dynamics of neural activity.

A hybrid entropy scheme was used to quantify source-signal complexity from complementary perspectives. Approximate, fuzzy, and permutation entropy were calculated and normalized separately because of their different numerical ranges. The normalized measures were then concatenated along the feature dimension to form a hybrid entropy vector for each frequency band and the corresponding original hybrid entropy feature matrix.

For the c-th ROI and r-th trial, the phase-space norm sequence was processed using the three entropy measures described in Section 2.3. After normalization, the values were concatenated into a three-dimensional hybrid entropy vector, whose components denote normalized approximate, fuzzy, and permutation entropy, respectively.

3.5 State-Space-Based Fused Feature Selection

After phase-space reconstruction and hybrid entropy extraction, the dynamical features were organized by ROI. To reduce the influence of inactive ROIs and redundant features, a state-space screening branch was constructed in parallel with the phase-space branch. This branch takes the preprocessed scalp EEG, lead-field matrix, and source-to-ROI mapping as inputs and identifies active ROIs through sparse source estimation and BIC-based model selection. The hybrid entropy features corresponding to the selected ROIs are retained and further ranked using ReliefF. The procedure is shown in figure 5.

As shown in figure 5, the procedure comprises state-space construction, row-sparse separation of the source output matrix, BIC-based model optimization, and active-ROI selection. The selected active ROIs are then used to constrain the hybrid entropy features, which are further ranked using ReliefF. Each step is described below.

3.5.1 State-Space Construction For a given head model, the lead-field matrix L describes the scalp potential distribution generated by unit source currents. Let the scalp EEG signal, cortical source

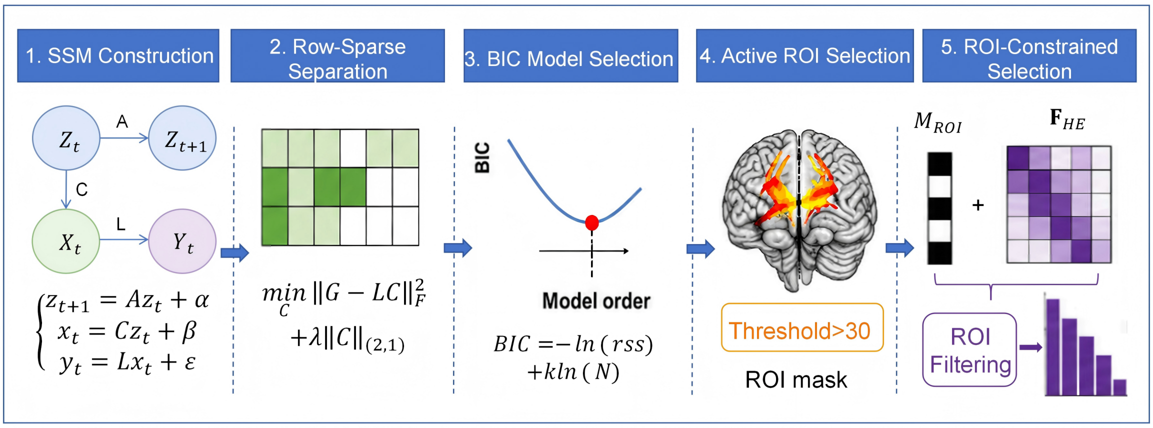


Figure 5. Detailed procedure for state-space-based ROI selection and ROI-constrained feature selection. (1) State-space model construction; (2) row-sparse separation of the source-output matrix; (3) BIC-based model selection; (4) active-ROI selection using the source-count threshold; and (5) ROI-constrained filtering and ReliefF ranking of hybrid entropy features.

signal, and measurement noise at time t be $y(t)$, $x(t)$, and $v(t)$, respectively. The forward model is expressed as

$$y(t) = Lx(t) + v(t). \tag{5}$$

where $y(t) \in \mathbb{R}^{N_c \times 1}$ denotes the EEG signal recorded from c channels, $L \in \mathbb{R}^{N_c \times N_s}$, is the lead-field matrix, s is the number of source locations, and $x(t) \in \mathbb{R}^{N_s \times 1}$ is the source-signal vector. This equation describes only the spatial mapping at a single time point and does not include the temporal dynamics of the signals.

To capture the temporal dynamics of source activity, a vector autoregressive (VAR) process was used to model the source signals[29]. The current source state $x(t)$ is represented as a linear combination of the source states at several preceding time points plus noise. Assuming a model order of n, the source dynamics are expressed as

$$x(t) = \sum_{k=1}^{n} M_k x(t-k) + \omega(t). \tag{6}$$

where M_k denotes the autoregressive coefficient matrix and $\omega(t)$ denotes process noise. Although the VAR representation incorporates temporal information, direct estimation of high-dimensional VAR parameters is computationally demanding and does not explicitly connect the source dynamics with the observed EEG signals.

To jointly describe system dynamics and the observation relationship, a state-space model with a latent neural state variable was constructed. The latent state $z(t)$ explicitly represents the internal dynamics of neural activity. A standard discrete-time state-space model contains a state equation and an output equation:

$$\begin{cases} z(t+1) = Az(t) + Bu(t) + \alpha, \\ x(t) = Cz(t) + Du(t) + \beta. \end{cases} \tag{7}$$

where $A \in \mathbb{R}^{m \times n}$ is the state-transition matrix describing the temporal evolution of the latent state; $C \in \mathbb{R}^{N_s \times n}$ is the state-space output matrix that maps the latent state to the observation space; $u(t)$ denotes the external input; and α and β denote state noise and observation noise, respectively.

In the EEG application considered here, brain activity was assumed to be internally generated without an external control input (i.e., $u(t) = 0$), and the observation target was the scalp EEG signal. Combining the forward model, the source signal $x(t)$ was regarded as a linear transformation of the latent state $z(t)$, namely $x(t) \approx Cz(t)$. Substituting this relationship into the forward equation establishes a direct mapping from the latent state to the scalp EEG. The resulting state-space model for EEG source localization is

$$\begin{cases} z(t+1) = Az(t) + \alpha, \\ x(t) = Cz(t) + \beta, \\ y(t) = LCz(t) + \varepsilon. \end{cases} \tag{8}$$

where ε denotes the composite observation noise. In this model, $L \cdot C$ forms the overall observation matrix from the latent state to the scalp EEG signal.

The composite matrix $G = L \cdot C$ in the observation equation was estimated as a whole. An initial system-matrix estimate was obtained by fitting the relationship between the state and output sequences using least squares. The resulting observation matrix $G \in \mathbb{R}^{c \times n}$ jointly contains lead-field and source-mapping information, where c is the number of EEG channels and n is the model order.

3.5.2 Row-Sparse Separation of Active Sources To separate the physiologically meaningful source output matrix C from the composite matrix G, the known lead-field matrix L was used to solve the linear equation $G = LC$. Because EEG source localization is ill-posed and active regions typically occupy only a small fraction of the cortex, a row-sparsity constraint was introduced, transforming the estimation of C into a convex optimization problem[30].

The objective function minimizes the Frobenius-norm reconstruction error with an additional row-sparse regularization term:

$$\min_{C} \ \frac{1}{2}\|G - LC\|_F^2 + \lambda \sum_{i=1}^{m} \|C_{i,:}\|_2. \tag{9}$$

where $\|\cdot\|_F$ denotes the Frobenius norm used to measure reconstruction error; $C_{i,:}$ is the i-th row of matrix C, corresponding to the i-th cortical source point; $\|\cdot\|_2$ is the L2 norm; λ is the regularization parameter controlling sparsity; and m is the total number of source points.

The objective encourages as many rows of matrix C as possible to become zero vectors while ensuring that $L \cdot C$ approximates G. In the estimated matrix $C \in \mathbb{R}^{m \times n}$, nonzero rows of C correspond to active source regions, whereas zero rows correspond to inactive regions. This sparsity suppresses the spatial spreading caused by volume conduction and separates active from inactive sources.

3.5.3 Model Optimization The regularization parameter λ and model order n directly affect source-estimation accuracy and model complexity. An excessively large λ produces an overly sparse model and may discard genuine active sources, whereas an excessively small λ yields a dense model and introduces noise. An improved Bayesian information criterion (BIC) was therefore used to adaptively select the optimal model.

The BIC is defined as

$$\mathrm{BIC} = -\ln(\mathrm{rss}) + k \ln(N). \tag{10}$$

where rss is the residual sum of squares, representing the fitting error between G and $L \cdot C$, namely $\mathrm{rss} = \|G - LC\|_F^2$; k is the model-complexity penalty, which is proportional to the number of nonzero rows in matrix C; and N is the number of data samples.

First, a series of λ values was uniformly sampled in the logarithmic interval $[10^{-10}, 0.1]$. For each λ, equation (9) was solved to obtain the corresponding matrix C, and the BIC value was calculated. The variation of BIC with λ was then analyzed. The desired model lies at the balance between fitting error and model complexity, corresponding to the minimum of the BIC curve. The λ^* that minimized BIC and its corresponding model order n^* were selected as the final model parameters. This procedure yields a parameter combination that balances fitting error and model complexity for subsequent active-source estimation.

3.5.4 ROI Selection To apply the state-space localization result to feature selection, a mapping between high-resolution source points and the ROI atlas was established. The ROI set is defined as $R = \{R_1, R_2, \ldots, R_M\}$, where $M = 28$ is the total number of ROIs. For each source point $s \in 1, 2, \ldots, S$, a mapping function $\phi(s)$ assigns it to a specific ROI R_j. If source point s lies within the anatomical extent of ROI R_j, then $\phi(s) = j$.

After BIC-based optimization, matrix C exhibits row sparsity. To quantify the activity level of each ROI, the number of active source points within each ROI was calculated. The activation indicator for source point s is defined as $I(s)$:

$$I(s) = \begin{cases} 1, & \|C_{s,:}\|_2 > 0, \\ 0, & \|C_{s,:}\|_2 = 0. \end{cases} \tag{11}$$

where $\|C_{s,:}\|_2$ denotes the L_2 norm of source point s, corresponding to the s-th row of matrix C. A source point was considered active when its norm was greater than zero. For the ROI R_j containing the source point, the number of active sources N_j within R_j was calculated.

$$N_j = \sum_{s=1}^{m} I(s), \qquad s \in R_j. \tag{12}$$

To remove weak activations caused by incidental activity or noise, a threshold of $\tau = 30$ was used. Only ROIs containing more than 30 active source points were retained; the remaining ROIs were treated as inactive and discarded.

After ROI-level screening, ReliefF was used to assess the importance of the remaining features[31]. ReliefF assigns feature weights by comparing within-class and between-class differences among neighboring samples. The 100 highest-ranked features were retained to form the final PSSF feature set for classification.

3.6 Evaluation Metrics and Classification Settings

To ensure comparable evaluation across feature representations, a linear support vector machine (SVM) was used as the common classifier. Performance was assessed using repeated five-fold cross-validation. In each repetition, all trials were randomly divided into five mutually exclusive subsets; four subsets were used for training and the remaining subset for testing, and the procedure continued until each subset had served as the test set once. The complete five-fold procedure was repeated ten times. Mean accuracy (Acc) and standard deviation (Std) were reported. Accuracy is defined as

$$\mathrm{Acc} = \frac{TP + TN}{TP + TN + FP + FN}. \tag{13}$$

The mean accuracy over the repeated cross-validation runs is calculated as

$$\mathrm{Avg} = \frac{1}{N} \sum_{i=1}^{N} \mathrm{Acc}_i. \tag{14}$$

The standard deviation quantifies the variability of accuracy across repeated evaluations and is defined as

$$\mathrm{Std} = \sqrt{\frac{1}{K-1} \sum_{k=1}^{K} (\mathrm{Acc}_k - \mathrm{Avg})^2}. \tag{15}$$

4 Results and Discussion

4.1 Classification Results on the Motor Imagery and Adult ADHD Datasets

To compare the discriminative ability of different feature representations, all experiments used the same linear SVM and evaluation settings. The comparison included power spectral density (PSD), individual complexity-entropy measures, the largest Lyapunov exponent (LLE)[32], and the PSSF feature set. The results are reported in tables 1 and 2.

As shown in table 1, the feature representations produced different classification performance for left- versus right-hand motor imagery across the nine participants in BCI Competition IV 2a. PSSF achieved the highest mean accuracy of 77.81%, exceeding PSD (66.20%) and the individual SampEn, PermEn, FuzzyEn, and LLE features. This result indicates that combining source-space constraints with complementary nonlinear dynamical descriptors provides more informative representations for motor imagery decoding.

At the participant level, PSSF showed stable performance for most participants and produced clear improvements over PSD for A04, A05, A06, and A09. For A05, accuracy increased from 58.03% with PSD to 88.19% with PSSF, corresponding to an improvement of 30.16 percentage points. The variation across participants also suggests that source-activity patterns and dynamical complexity differ across individuals, with fused features providing a larger benefit for some participants than for others.

Table 1. SVM classification results for different features on the BCI dataset.

Subject	PSD	SampEn	PermEn	FuzzyEn	LLE	PSSF
A01	69.26	74.36	74.03	64.71	61.31	70.98
A02	51.34	55.47	53.07	51.58	49.47	71.81
A03	83.29	85.64	92.30	60.42	70.42	82.80
A04	63.34	56.59	64.04	60.63	56.99	78.25
A05	58.03	58.27	60.53	77.36	51.06	88.19
A06	57.68	51.24	52.04	64.72	52.58	74.87
A07	63.06	55.80	61.60	75.67	60.68	68.64
A08	87.26	85.57	93.24	86.19	86.65	90.88
A09	62.57	59.33	74.44	62.72	57.57	73.84
Avg	66.20	64.70	69.48	67.11	60.75	77.81
Std	11.24	12.69	14.44	10.01	10.94	7.41

LLE primarily characterizes the local divergence of phase-space trajectories and achieved a mean accuracy of 60.75%, lower than most entropy-based features. Approximate, fuzzy, and permutation entropy describe complementary properties related to pattern persistence, fuzzy similarity, and ordinal structure. By integrating these dynamical properties with active-ROI constraints, PSSF achieved a higher mean classification accuracy.

Table 2. SVM classification results for different features on the self-collected adult ADHD dataset.

Pair	PSD	SampEn	PermEn	FuzzyEn	LLE	PSSF
A01	57.34	84.12	94.51	99.06	71.49	99.33
A02	59.61	86.05	92.58	97.82	71.33	99.47
A03	59.39	73.40	81.24	96.72	58.75	96.46
A04	68.67	84.07	98.06	98.59	67.24	98.13
Avg	61.25	81.91	91.59	98.05	67.20	98.35
Std	4.37	4.98	6.30	0.89	5.17	1.21

Table 2 reports the results for ADHD–HC participant pairs (A01–A04) from the self-collected adult ADHD dataset. PSSF achieved the highest mean accuracy of 98.35%, while FuzzyEn also produced a high mean accuracy of 98.05%. These results indicate that phase-space-based complexity representations capture dynamical differences between ADHD and healthy control samples, and that state-space-based ROI screening combined with hybrid entropy provides an informative overall feature representation.

4.2 Ablation Study of Phase-Space Reconstruction and State-Space Modeling

To examine the effect of jointly introducing phase-space reconstruction and state-space modeling, classification performance on the adult ADHD dataset was compared between conditions without PSR and SSM and with both PSR and SSM. The results are shown in table 3.

Without PSR and SSM, the mean accuracies of SampEn, PermEn, FuzzyEn, and HybridEn were 90.96%, 90.96%, 91.07%, and 92.63%, respectively. With PSR and SSM, the mean accuracies of PermEn, FuzzyEn, and HybridEn reached 91.59%, 98.05%, and 98.35%, respectively; FuzzyEn and HybridEn improved by 6.98 and 5.72 percentage points. SampEn did not improve under this setting, indicating that different complexity measures respond differently to the phase-space transformation. Overall, the results suggest that the joint use of PSR and state-space-based ROI screening improves the performance of selected complexity features, particularly FuzzyEn and HybridEn.

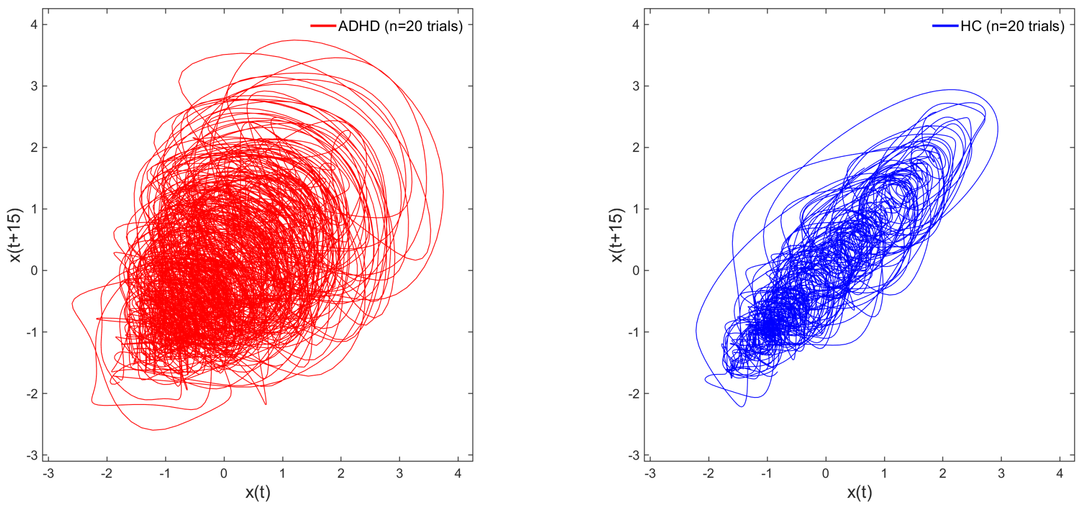
4.3 Visualization of Phase-Space Trajectories

To illustrate how PSR characterizes the dynamics of source time series, phase-space trajectories were visualized using the adult ADHD dataset. ROI-level source time series were embedded with time delays, and a representative combination was selected by considering the paired dataset, ROI, time delay, and embedding dimension. ROI11, a time delay of 15, and an embedding dimension of 2 were used to visualize representative trials from the healthy control and ADHD groups. This analysis was used only to examine differences in source-level dynamical trajectories and was not involved in classifier training or performance calculation. figure 6 presents the two-dimensional trajectories using identical coordinate ranges for the two groups. The healthy control trajectories were relatively concentrated and mainly extended along an oblique elliptical region, whereas the ADHD trajectories

Table 3. Ablation results for the effects of phase-space reconstruction and state-space modeling on classification performance (%).

	Without PSR and SSM				With PSR and SSM			
Pair	SampEn	PermEn	FuzzyEn	HybridEn	SampEn	PermEn	FuzzyEn	HybridEn
A01	94.20	95.52	90.60	94.39	84.12	94.51	99.06	99.33
A02	94.89	84.90	94.21	91.56	86.05	92.58	97.82	99.47
A03	82.35	86.65	87.53	87.89	73.40	81.24	96.72	96.46
A04	92.41	96.78	91.93	96.67	84.07	98.06	98.59	98.13
Avg	90.96	90.96	91.07	92.63	81.91	91.59	98.05	98.35
Std	5.05	5.24	2.42	3.28	4.98	6.30	0.89	1.21

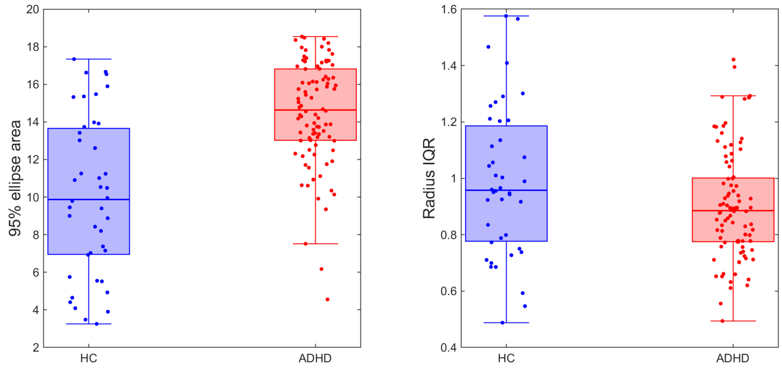
covered a broader area and showed a more expanded outer envelope. These differences in compactness and coverage provide an intuitive illustration of source-level dynamical differences represented by PSR.



(a) ADHD **(b) Healthy control**

Figure 6. Representative phase-space trajectories in the adult ADHD dataset. (a) ADHD and (b) healthy-control trajectories for ROI11, with a time delay of 15 and an embedding dimension of 2. The two panels use identical coordinate ranges.

To further describe these trajectory differences, phase-space geometric indices were calculated and compared between groups, as shown in figure 7. The 95% ellipse area represents the overall coverage of a two-dimensional trajectory, whereas the radius interquartile range (Radius IQR) measures the dispersion of trajectory-point distances from the center. The ADHD group exhibited a larger overall 95% ellipse area than the healthy control group, consistent with the expanded envelope in figure 6 and indicating a broader range of state evolution in the selected source region. Radius IQR did not show a corresponding increase, suggesting that the group difference was not primarily attributable to uniform dispersion around the center. Taken together, the ADHD pattern was characterized by an expanded outer trajectory envelope and a wider range of state evolution rather than a general increase in local dispersion.



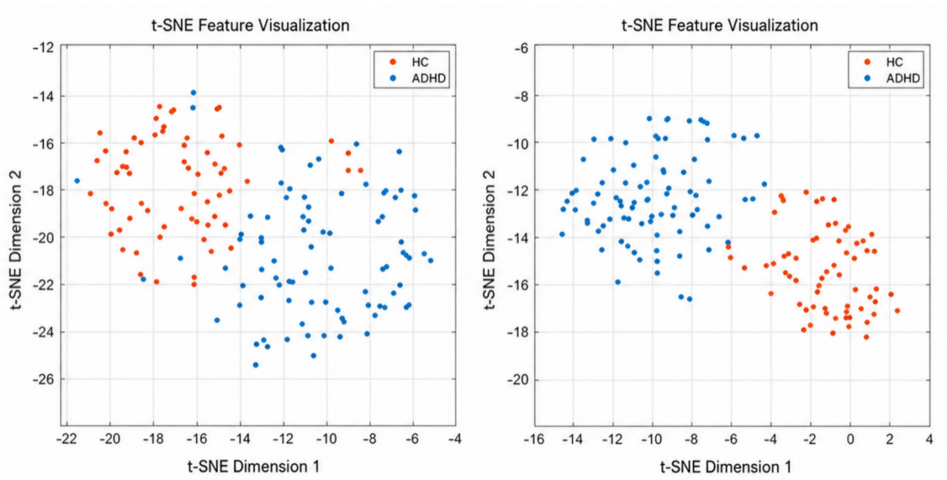
(a) 95% ellipse area **(b) Radius IQR**

Figure 7. Between-group distributions of phase-space geometric indices in the adult ADHD dataset. (a) The 95% ellipse area represents the overall coverage of the reconstructed phase-space trajectory. (b) The radius interquartile range (Radius IQR) represents the dispersion of trajectory-point distances from the trajectory center. The boxplots summarize the distributions of the corresponding geometric indices for the ADHD and healthy-control groups.

Taken together, the trajectory patterns in figure 6 and the geometric indices in figure 7 show that PSR presents differences between ADHD and healthy control samples at the level of source-region dynamics and provides a visual basis for interpreting group differences in source-activity evolution.

4.4 t-SNE Visualization of Feature Distributions

To visually compare feature distributions before and after PSR and state-space modeling, t-SNE was used to project the high-dimensional features into two dimensions. As shown in figure 8, most samples already exhibited a degree of class clustering before processing, although local overlap remained. After PSR and state-space modeling, the two classes formed clearer clusters in the two-dimensional projection and showed less mixing near the class boundary. The visualization therefore illustrates changes in the distributional structure of the features after transformation.



(a) Before PSR and SSM **(b) After PSR and SSM**

Figure 8. t-SNE visualization of feature distributions before and after phase-space reconstruction and state-space-based ROI screening. (a) Feature distributions before PSR and SSM processing and (b) feature distributions after PSR and SSM processing. The same class labels and visualization settings were used for both panels.

The t-SNE patterns are consistent with the overall trends of the classification and ablation results, with the processed features showing a clearer class structure in the low-dimensional projection. Because t-SNE is used here as a qualitative visualization, it is not treated as a direct measure of classification performance or generalization.

4.5 Functional Correlation Analysis of ROIs Selected by the State-Space Model

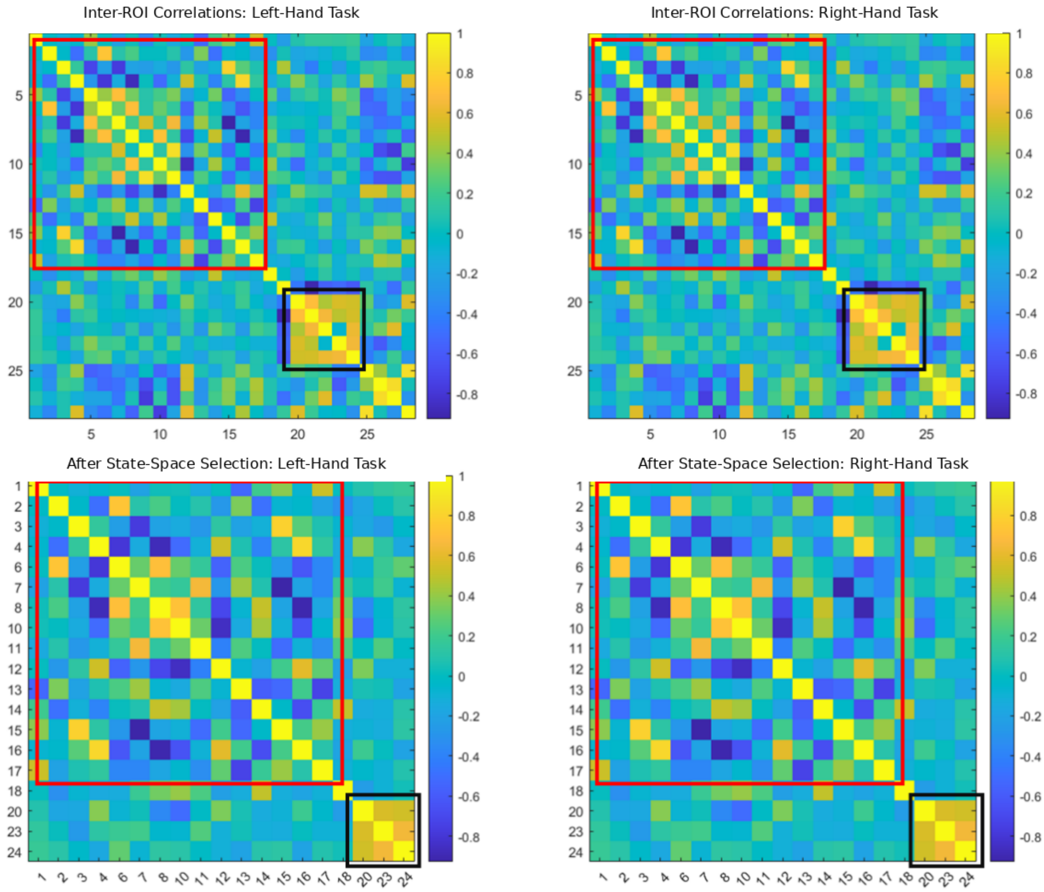


Figure 9. Inter-ROI correlation matrices for the left- and right-hand motor imagery tasks before and after state-space-based ROI selection. The upper-left and upper-right panels show the correlation matrices for the left- and right-hand tasks before selection, respectively. The lower-left and lower-right panels show the corresponding matrices after state-space-based ROI selection. The red boxes highlight the main correlation patterns among the selected sensorimotor-related ROIs, whereas the black boxes indicate a compact correlation pattern among the posterior ROIs. The color scale represents the Pearson correlation coefficient.

To examine inter-ROI coordination before and after state-space-based ROI selection and to interpret the spatial relevance of the selected ROIs, Pearson correlation coefficients were calculated between ROIs in the BCI dataset. As shown in figure 9, before selection, both the left- and right-hand motor imagery tasks exhibited broad correlation patterns across multiple ROIs. Strong block-like patterns were observed among ROI indices 5–10, extending in some cases across indices 2–17, and local high-correlation patterns were also present around indices 20–25. These results indicate that the original source time series contain widespread inter-ROI coordination, including both task-related and potentially redundant correlation patterns.

After state-space-based ROI selection, the inter-ROI correlation structure became more concentrated. Correlation patterns involving ROI index 19 and indices 25–28 were attenuated, whereas the retained relationships were mainly distributed among sensorimotor-related ROIs, including BA3a, BA3b, and BA4a around the central sulcus. These regions are associated with somatosensory processing, motor execution, and motor imagery. The results suggest that state-space-based ROI selection reduces redundant spatial correlations while retaining task-related patterns, thereby supporting the neurophysiological plausibility of the selected ROIs.

5 Conclusion

This study proposed PSSF, a phase-space and state-space fusion method for feature decoding after EEG source localization. PSSF combines nonlinear dynamical representations of source time series with active-ROI constraints. On the BCI Competition IV 2a motor imagery dataset and the self-collected adult ADHD dataset, PSSF achieved mean classification accuracies of 77.81% and 98.35%, respectively. The ablation results indicate that the joint use of PSR and state-space-based

ROI screening improves the performance of features such as FuzzyEn and HybridEn. Phase-space trajectories, t-SNE distributions, and inter-ROI correlation patterns further illustrate between-class dynamical differences and task-related spatial organization. These results suggest that PSSF provides a unified method for combining source-space constraints with nonlinear dynamical information and supports the neurophysiological interpretation of the resulting features. Several limitations remain. The time delay and embedding dimension for PSR were determined within empirically defined search ranges, and iterative state-space estimation and high-dimensional feature fusion introduce substantial computational cost. In addition, the self-collected dataset was limited in scale, and independent cross-participant validation was not conducted; generalization to unseen participants therefore requires further evaluation on larger cohorts. Future work will investigate adaptive parameter optimization, lightweight state inference, and participant-independent validation to improve real-time performance and generalizability.

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The authors declare no competing interests.

Data availability
The BCI Competition IV 2a dataset is publicly available from the original competition repository. The self-collected adult ADHD dataset is not publicly available because it contains human-participant data and is subject to institutional ethical and privacy restrictions.

References

[1] Iqbal, A., Suhag, A. K., Kumar, N., & Bhardwaj, A. (2025). Use of BCI systems in the analysis of EEG signals for motor and speech imagery task: A SLR. ACM Computing Surveys, 58(2), 1–25.

[2] Feng, Z., Kakkos, I., Matsopoulos, G. K., Guan, C., & Sun, Y. (2025). Explaining E/MEG source imaging and beyond: An updated review. IEEE Journal of Biomedical and Health Informatics, 29(12), 9271–9286.

[3] Qin, X., Du, L., Jiao, X., et al. (2023). Evaluation of brain source localization methods based on test-retest reliability with multiple session EEG data. IEEE Transactions on Biomedical Engineering, 70(7), 2080–2090.

[4] Jiao, M., Yang, S., Xian, X., et al. (2024). Multi-modal electrophysiological source imaging with attention neural networks based on deep fusion of EEG and MEG. IEEE Transactions on Neural Systems and Rehabilitation Engineering, 32, 2492–2502.

[5] Hamid, L., Habboush, N., Stern, P., et al. (2021). Source imaging of deep-brain activity using the regional spatiotemporal Kalman filter. Computer Methods and Programs in Biomedicine, 200, 105830.

[6] Takens, F. (1981). Detecting strange attractors in turbulence. Lecture Notes in Mathematics, 898, 366–381.

[7] Chen, M., Fang, Y., & Zheng, X. (2014). Phase space reconstruction for improving the classification of single trial EEG. Biomedical Signal Processing and Control, 11, 10–16.

[8] Dovedi, T., Upadhyay, R., & Kumar, V. (2023). Multivariate variational mode decomposition & phase space reconstruction based motor imagery EEG classification. Computers and Electrical Engineering, 108, 108737.

[9] Kumar, S. T. S., & Kasthuri, N. (2019). EEG seizure classification based on exploiting phase space reconstruction and extreme learning. Cluster Computing, 22(5), 11477–11487.

[10] Skaria, S., & Savithriamma, S. K. (2024). Automatic classification of seizure and seizure-free EEG signals based on phase space reconstruction features. Journal of Biological Physics, 50(2), 181–196.

[11] Lu, J., Li, A., Li, K., et al. (2024). An EEG study on β–γ phase-amplitude coupling-based functional brain network in epilepsy patients. IEEE Journal of Biomedical and Health Informatics, 28(6), 3446–3456.

[12] Wang, Z., et al. (2024). Automated seizure detection based on state-space model identification. Sensors, 24(6), 1902.

[13] Shen, H., Xu, M., Guez, A., et al. (2019). An accurate sleep stages classification method based on state space model. IEEE Access, 7, 125268–125279.

[14] Cheung, B. L. P., Riedner, B. A., Tononi, G., et al. (2010). Estimation of cortical connectivity from EEG using state-space models. IEEE Transactions on Biomedical Engineering, 57(9), 2122–2134.

[15] Carrara, I., Aristimunha, B., Corsi, M. C., de Camargo, R. Y., Chevallier, S., & Papadopoulo, T. (2025). Geometric neural network based on phase space for BCI-EEG decoding. Journal of Neural Engineering, 22(1), 016049.

[16] Al Fahoum, A., & Zyout, A. (2024). Wavelet transform, reconstructed phase space, and deep learning neural networks for EEG-based schizophrenia detection. International Journal of Neural Systems, 34(09), 2450046.

[17] Gramfort, A., Papadopoulo, T., Olivi, E., & Clerc, M. (2010). OpenMEEG: Opensource software for quasistatic bioelectromagnetics. BioMedical Engineering OnLine, 9, 45.

[18] Pascual-Marqui, R. D. (2002). Standardized low-resolution brain electromagnetic tomography (sLORETA): Technical details. Methods and Findings in Experimental and Clinical Pharmacology, 24(Suppl D), 5-12.

[19] Sadat-Nejad, Y., & Beheshti, S. (2021). Efficient high resolution sLORETA in brain source localization. Journal of Neural Engineering, 18(1), 016012.

[20] Pincus, S. M. (1991). Approximate entropy as a measure of system complexity. Proceedings of the National Academy of Sciences, 88(6), 2297–2301.

[21] Bandt, C., & Pompe, B. (2002). Permutation entropy: A natural complexity measure for time series. Physical Review Letters, 88(17), 174102.

[22] Chen, W., Wang, Z., Xie, H., & Yu, W. (2007). Characterization of surface EMG signal based on fuzzy entropy. IEEE Transactions on Neural Systems and Rehabilitation Engineering, 15(2), 266–272.

[23] Tangermann, M., Müller, K. R., Aertsen, A., et al. (2012). Review of the BCI Competition IV. Frontiers in Neuroscience, 6, 55.

[24] Tadel, F., Baillet, S., Mosher, J. C., Pantazis, D., & Leahy, R. M. (2011). Brainstorm: A user-friendly application for MEG/EEG analysis. Computational Intelligence and Neuroscience, 2011, 879716.

[25] Bi, J., et al. (2024). Classification of motor imagery using chaotic entropy based on sub-band EEG source localization. Journal of Neural Engineering, 21(3), 036016.

[26] Fraser, A. M., & Swinney, H. L. (1986). Independent coordinates for strange attractors from mutual information. Physical Review A, 33(2), 1134–1140.

[27] Kennel, M. B., Brown, R., & Abarbanel, H. D. I. (1992). Determining embedding dimension for phase-space reconstruction using a geometrical construction. Physical Review A, 45(6), 3403–3411.

[28] Richman, J. S., & Moorman, J. R. (2000). Physiological time-series analysis using approximate entropy and sample entropy. American Journal of Physiology-Heart and Circulatory Physiology, 278(6), H2039–H2049.

[29] Manomaisaowapak, P., Nartkulpat, A., & Songsiri, J. (2022). Granger causality inference in EEG source connectivity analysis: A state-space approach. IEEE Transactions on Neural Networks and Learning Systems, 33(7), 3146–3156.

[30] Haufe, S., Müller, K. R., Nolte, G., & Krämer, N. (2010). Modeling sparse connectivity between underlying brain sources for EEG/MEG. IEEE Transactions on Biomedical Engineering, 57(8), 1954–1963.

[31] Robnik-Šikonja, M., & Kononenko, I. (2003). Theoretical and empirical analysis of ReliefF and RReliefF. Machine Learning, 53(1), 23–69.

[32] Nassehi F, Sönmez İ, Sabanoglu B, et al. Detection of Attention Deficit and Hyperactivity Disorder by Nonlinear EEG Analysis[C]//2022 Medical Technologies Congress (TIPTEKNO). IEEE, 2022: 1-4.